

Analyzing Pedestrian Accessibility to Parks and Green Spaces in Chicago City

Course: AGOG 597 - Advanced GIS
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THE IMPORTANCE OF URBAN GREEN SPACES

Access to parks and green spaces is fundamental to urban quality of life. Research consistently shows that nearby parks provide critical benefits:

- **Public Health:** Regular park access is linked to increased physical activity, reduced obesity rates, and improved mental health outcomes
- **Social Equity:** Low-income and minority neighborhoods often have less access to quality green space - a key environmental justice issue
- **Urban Planning Standard:** The "10-minute walk rule" suggests every resident should live within a 10-minute walk (~800 meters) of a park or green space

STUDY AREA:

Chicago, Illinois:

Population: 2,721,326

Area: 606 km² (234 square miles)



Chicago, with 2.7 million residents, faces a critical question: Are these parks truly accessible to the people who need them?

Traditional accessibility studies use simple distance measurements - but do these capture the reality of urban life? When streets, buildings, and the Chicago River stand between residents and parks, straight-line distances may tell a misleading story.

RESEARCH QUESTION

Main Question: How many Chicago residents have pedestrian access to parks within walking distance?

- What percentage of Chicago's land area lies within 400m and 800m of parks?

METHODOLOGY

Data Acquisition

Park boundaries, city boundary, street centerlines (Chicago Data Portal, 2024)

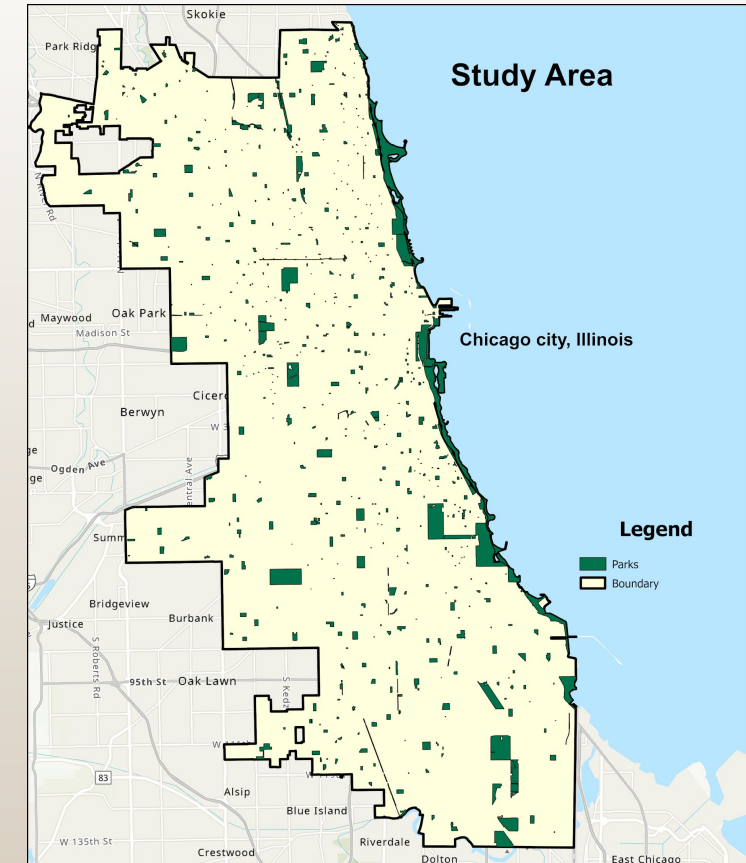
Population data (US Census ACS, 2024)

All data projected to NAD 1983 UTM Zone 16N

Euclidean Buffer Analysis: 400m and 800m buffers around parks representing straight-line walking distances

Network Service Area Analysis: 800m service areas along actual street network accounting for real walking constraints

Comparison: Area coverage and population accessibility between both methods



RESULTS - OVERVIEW

Summary of Findings

Buffer Analysis Findings

Traditional straight-line distance measurements:

Method	Distance	Area Coverage	Population Served
Euclidean Buffer	400m	5.77%	~156,973 people
Euclidean Buffer	800m	9.22%	~250,880 people
Network-Based	800m	5.78%	~157,230 people

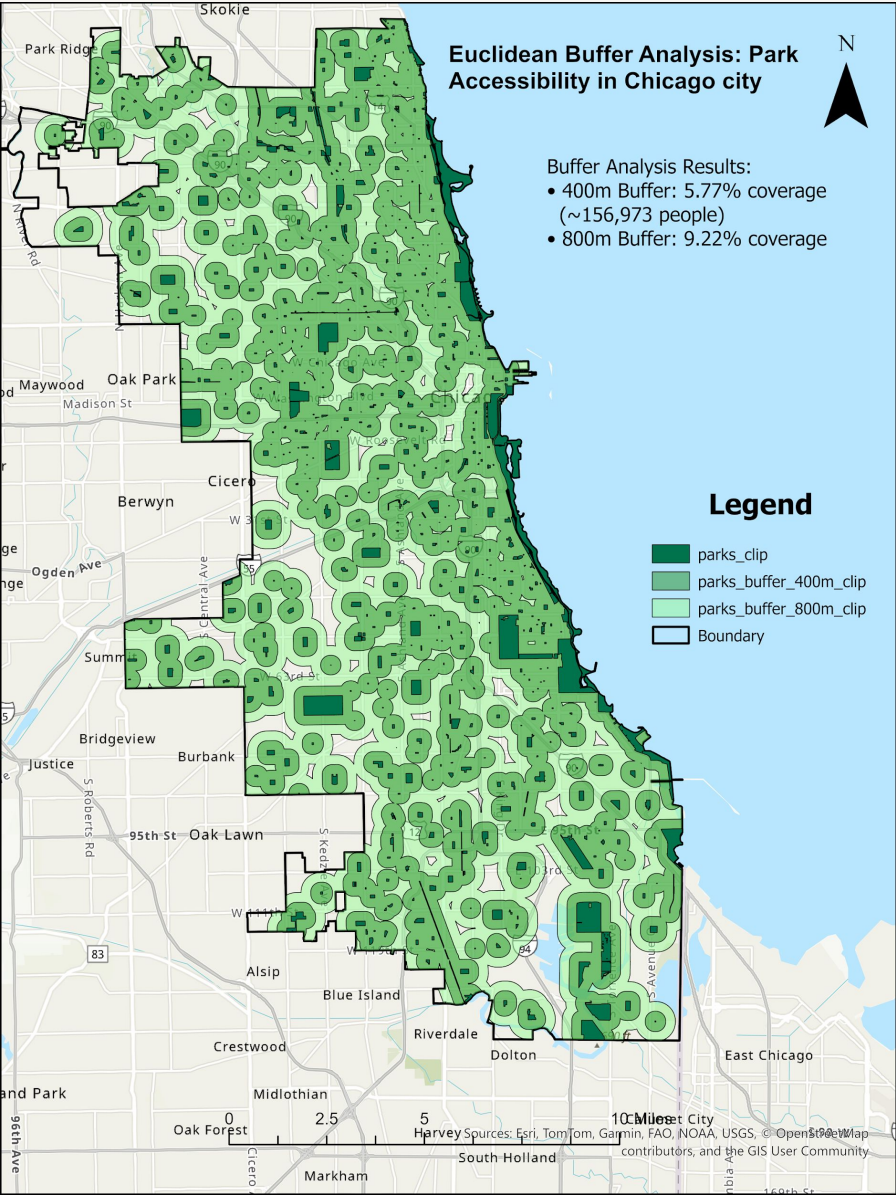
Interpretation:

400m Buffer (5-minute walk):

- Covers only 5.77% of Chicago's total area
- Approximately 157,000 residents live within this distance
- Represents immediate neighborhood access

800m Buffer (10-minute walk - Planning Standard):

- Covers 9.22% of Chicago's total area
- Approximately 251,000 residents served
- Still means over 90% of Chicago residents lack access within the standard planning threshold



RESULTS - OVERVIEW

Network-Based Analysis Findings
Realistic walking distance along actual street network:

Method	Distance	Area Coverage	Population Served
Network Service Area	800m walking	5.78%	~157,230 people

Network-based 800m ≈ Euclidean 400m

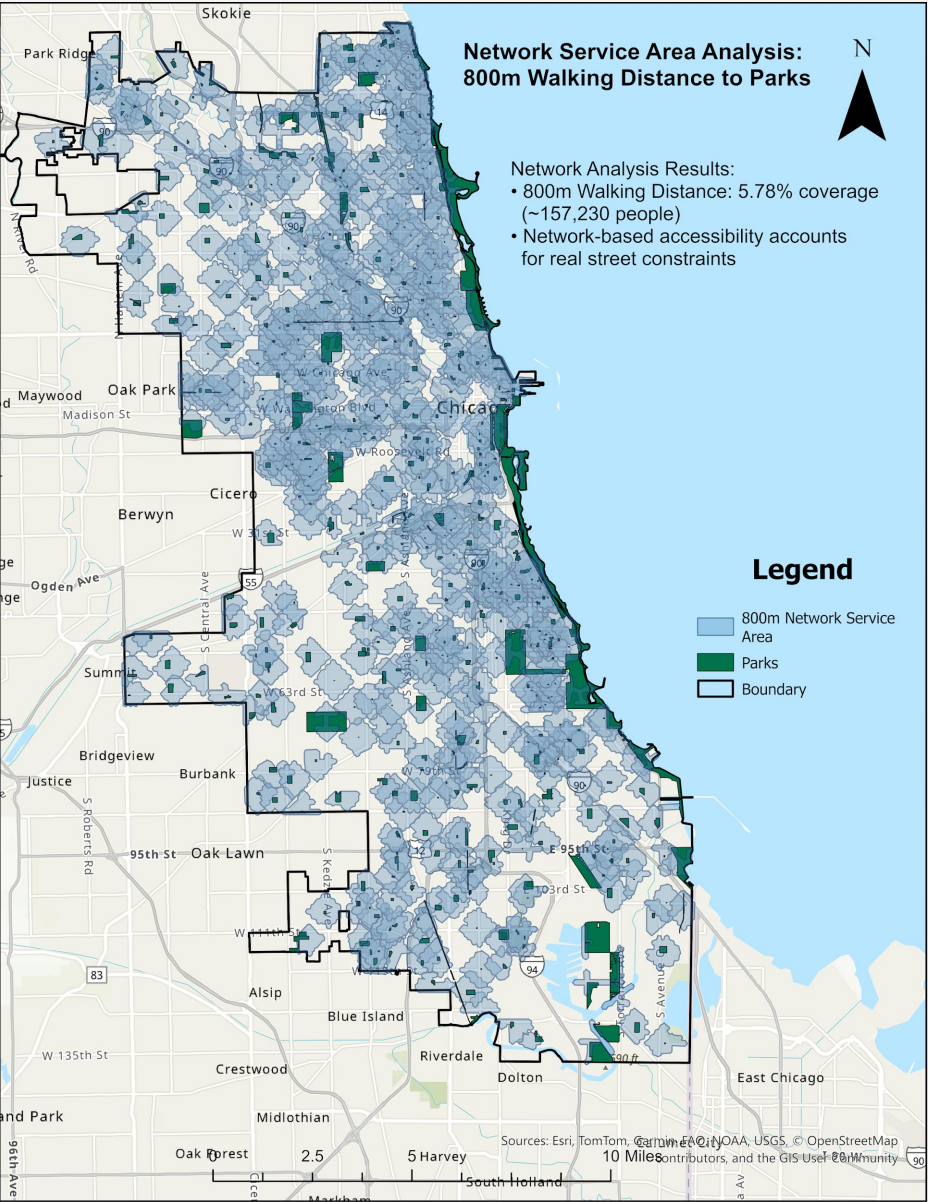
The network-based 800-meter walking distance provides nearly identical coverage to the 400-meter straight-line buffer.

What This Means:

- Walking 800m along streets provides the same coverage as 400m straight-line distance
- The urban environment doubles the effective distance pedestrians must travel
- Street network constraints reduce accessibility by approximately 50%

Key Insight:

While the Euclidean 800m buffer suggested 9.22% coverage (~251,000 people), the reality is only 5.78% coverage (~157,000 people) when accounting for actual walking routes. Over 94,000 residents were incorrectly classified as having park access.



RESULTS - DIRECT COMPARISON

Side-by-Side Comparison (800m Standard)

Metric	Euclidean Buffer	Network-Based	Difference
Area Coverage	9.22%	5.78%	-3.44%
Population Served	~250,880	~157,230	-93,650
Overestimation	—	—	+59%

Visual Analysis:

- Green areas: Covered by Euclidean buffer only
- Blue areas: Covered by network analysis
- Blue-green overlap: Accessible by both methods

Geographic Patterns:

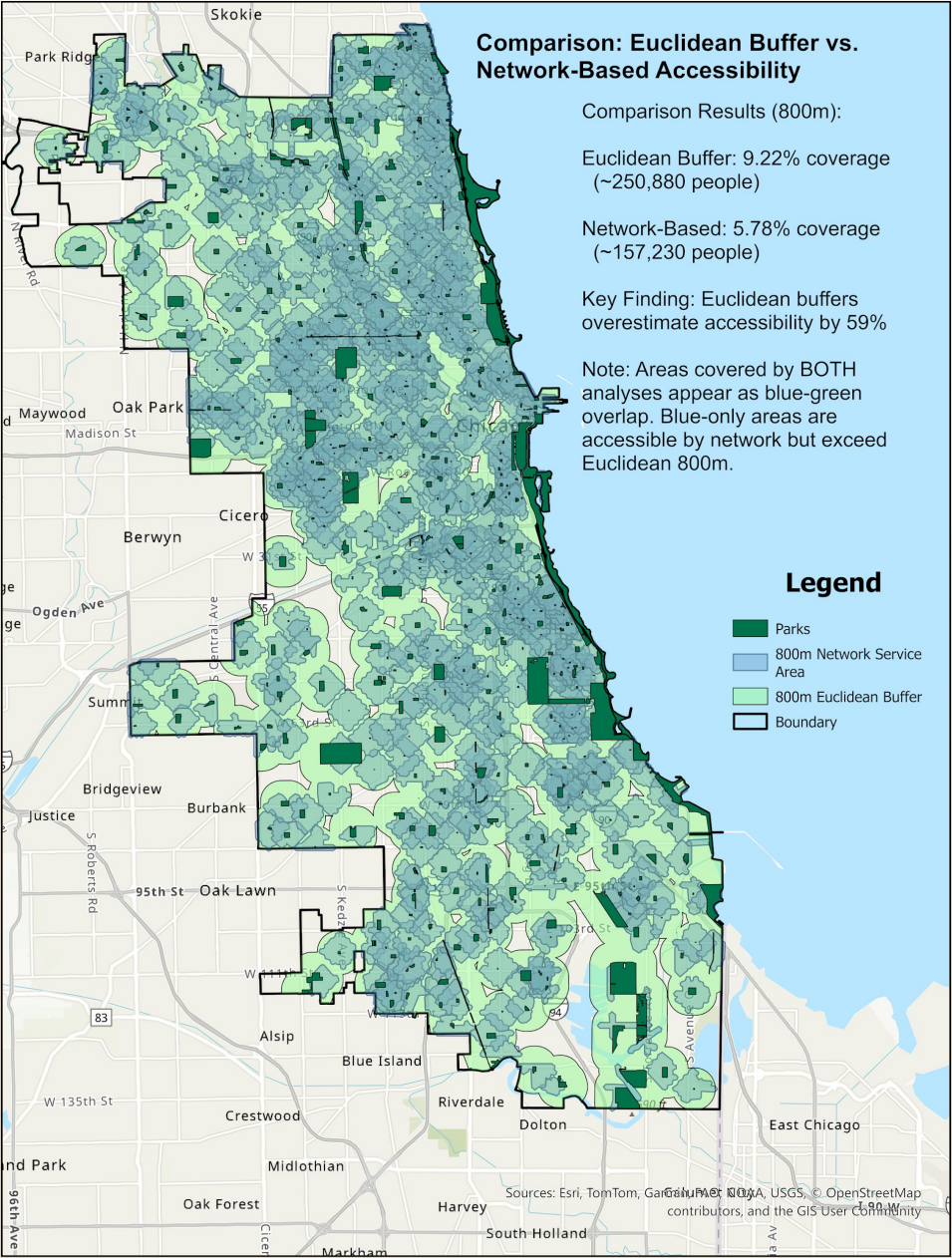
Areas with largest discrepancies occur where:

- Street grid is irregular or interrupted
- Physical barriers exist (Chicago River, railways, highways)
- Park access points are limited
- Street connectivity is poor (dead-ends, gated areas)

The Bottom Line:

Traditional Euclidean buffer analysis overestimates park accessibility by 59%, potentially leading to:

- Misidentification of well-served neighborhoods
- Underinvestment in areas that truly lack access
- Inequitable distribution of park resources



DISCUSSION

Factors Contributing to the 59% Overestimation:

1. Urban Grid System (Manhattan Distance Effect)

- Chicago's orthogonal street grid prevents diagonal movement
- Walking distance increases by ~41% compared to straight-line
- Example: 800m straight-line = ~1,130m actual walking distance

2. Physical Barriers

- Chicago River: Limited bridge crossings constrain north-south movement
- Lake Michigan: Eastern boundary blocks access
- Highways & Railways: I-90/94, rail corridors fragment neighborhoods
- Private Property: Buildings and gated areas block through-routes

3. Street Network Characteristics

- Dead-end streets requiring backtracking
- Limited park entrances - not all park edges are accessible
- One-way streets and traffic restrictions
- Discontinuous street patterns in some neighborhoods

4. Scale of Impact

- Central areas: Relatively good street connectivity, smaller discrepancy
- Peripheral areas: Irregular grids, larger discrepancy between methods
- River-adjacent neighborhoods: Most significant access limitations

Real-World Example:

A resident living 750m straight-line from a park may actually need to walk 1,200m following streets, exceeding the 800m threshold and losing "access" status.

CONCLUSIONS

Key Findings:

1. Limited True Accessibility

- Only 5.78% of Chicago has genuine pedestrian access to parks within 800m walking distance
- Approximately 157,000 residents (5.8% of population) can reach parks within a 10-minute walk
- Over 2.5 million residents (94%) live beyond the standard accessibility threshold

2. Methodological Gap

- Euclidean buffers overestimate accessibility by 59%
- Traditional methods misclassify nearly 94,000 residents as having park access
- Straight-line distance measures are inadequate for pedestrian accessibility assessment

3. Urban Context Matters

- Street network configuration significantly constrains accessibility
- Physical barriers (rivers, highways, buildings) reduce effective access
- Urban form doubles the effective distance pedestrians must travel

4. Planning Implications

- Network-based analysis is essential for accurate accessibility measurement
- Current planning decisions may be based on overoptimistic assumptions
- Environmental justice requires realistic assessment of who has actual access

Broader Significance:

This research demonstrates that analytical methods matter. The choice between Euclidean and network-based approaches can lead to fundamentally different conclusions about urban equity and service provision.

RECOMMENDATIONS

For Urban Planning Practice:

- Adopt network-based analysis as the standard for all pedestrian accessibility studies
- Re-evaluate existing park service assessments using network methods
- Identify priority areas for new park development based on network distance gaps
- Consider street connectivity when siting new parks - not just geographic location

Policy Implications:

- Update park planning standards to specify network-based distance measurements
- Integrate accessibility analysis into equity and environmental justice assessments
- Coordinate park placement with pedestrian infrastructure improvements (sidewalks, crossings)

For Future Research

Enhanced Analysis:

- Incorporate census tract-level population data for more precise demographic analysis
- Include park quality and amenity variables (not all parks provide equal value)
- Analyze accessibility by demographic groups (age, income, race/ethnicity) for equity assessment
- Account for seasonal variations (winter accessibility in Chicago differs significantly)

Additional Factors:

- Sidewalk quality and completeness - not all streets are pedestrian-friendly
- Street crossing difficulty - major arterials create psychological barriers
- Safety perceptions - crime rates and lighting affect willingness to walk
- Topography and elevation changes (minimal in Chicago but significant elsewhere)
- Park entrance locations - actual access points, not entire park perimeter

THANKYOU